Prediction Market Optimization

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Problem Statement - Why Prediction Market?

- Prediction markets are a real-money, real-time setting where beliefs update quickly
- Categories (sports, elections, geopolitics, macro, etc.) create a natural cross-domain diversification problem
- Especially, prediction markets also accurately reflect market sentiment towards topics including macro, finance, election, etc.
- **Core proposal idea: explicitly penalize over-exposure to a single event domain to reduce tail risk**
- A naive equal-weight portfolio is simple but may miss structure in return/risk dynamics
- A constrained online optimizer can adapt over time while respecting diversification limits

Process Overview

1. Ingest active events
2. Flatten + domain tagging
   - Convert event->market nested JSON to market rows, map tags to category/domain, and keep YES token IDs for binary interpretation.
3. Category-balanced market selection
   - Prioritize high-liquidity categories, apply per-category caps
4. Fetch token histories
5. Build cashed Week 8 dataset
6. Baseline run
   - Build price matrix, compute returns, allocate equal weight by domain, and save baseline metrics/time series.
7. Constrained run + running
8. Holdout evaluation + diagnostics

Core Math Formulas (1/2) - Definitions

- $r_t \in R^N$: vector of token returns at step t
- $w_t \in \Delta^N$: portfolio weights on simplex, $\sum_i w_{t,i} = 1$ and $w_{t,i} \geq 0$
- D: set of event domains/categories
- $S_d(w) = \sum_{i \in d} w_i$: total capital allocated to domain d
- L_d: domain limit for domain d

Core Math Formulas (2/2) - Concepts

Step return:

$$R_t = w_t^{\mathsf{T}} r_t$$

Cumulative value:

$$V_t = \prod_{s=1}^{t} (1 + R_s)$$

Drawdown:

$$\mathrm{DD}_t = \frac{V_t}{\max_{s \le t} V_s} - 1$$

Sortino:

$$\mathrm{Sortino}(R) = \frac{E[R]}{\sqrt{E!\left[\left(\min(R, 0)\right)^2\right]} + \varepsilon}$$

Domain overexposure penalty:

$$P_{\mathrm{domain}}(w) = \sum_{d \in D} \max!\left(0, S_d(w) - L_d\right)^2$$

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Concentration penalty:

$$P_{\mathrm{conc}}(w) = \sum_{i} \max\left(0, w_i - w_{\max}\right)^2$$

Entropy bonus:

$$B_{\mathrm{ent}}(w) = -\sum_{i} w_i \log(w_i + \varepsilon)$$

Weight mixing with uniform allocation u:

$$\tilde{w}_t = (1 - \alpha), \mathrm{softmax}(z_t) + \alpha, u$$

Objective per update window:

$$\max\ \left[\mathrm{Sortino}(R_{\mathrm{window}}) - \lambda, P_{\mathrm{domain}}(\tilde{w}_t) - \lambda_c, P_{\mathrm{conc}}(\tilde{w}_t) + \lambda_e, B_{\mathrm{ent}}(\tilde{w}_t)\right]$$

Proposal baseline objective (special case framing):

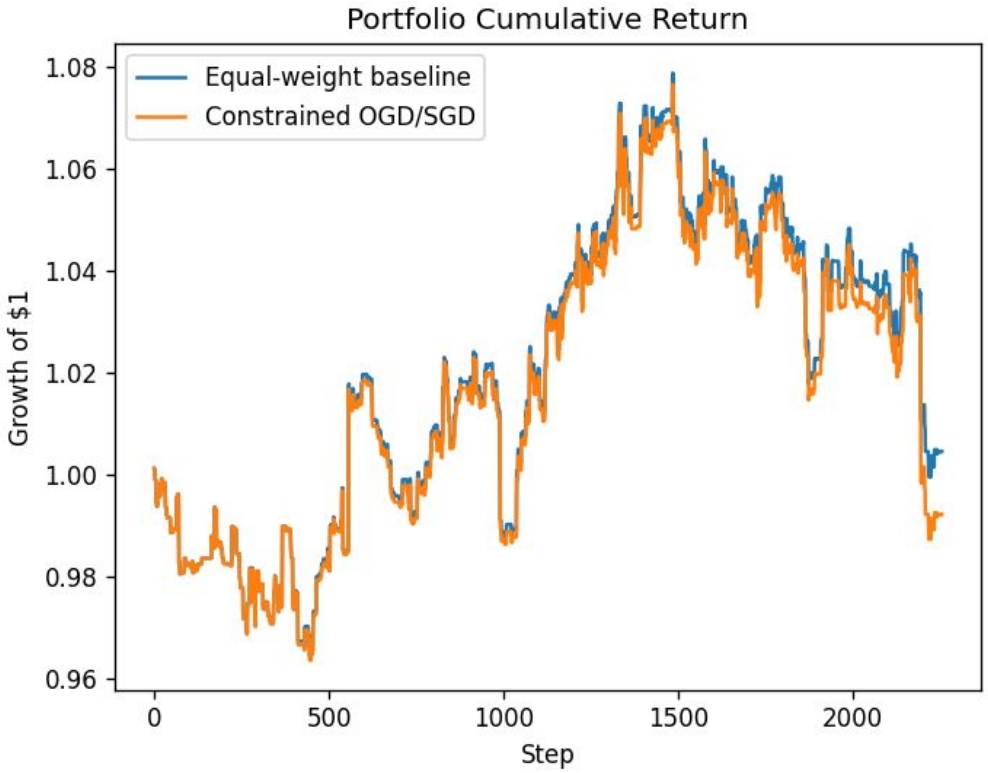
$$\max\ \mathrm{Sortino}_t(R_p) - \lambda \sum_{d \in D} \left[\max\left(0, S_d - L_d\right)\right]^2$$

Evaluation Design

- Walk-forward tuning: Hyperparameters selected using rolling in-sample windows to maximize penalized Sortino ratio.
- Holdout testing: Final performance evaluated strictly out-of-sample on unseen data.
- Baseline comparison: Equal-weight portfolio used as benchmark for return, volatility, and drawdown.
- Metrics tracked: Cumulative return, max drawdown, volatility, rolling mean return, and return distribution

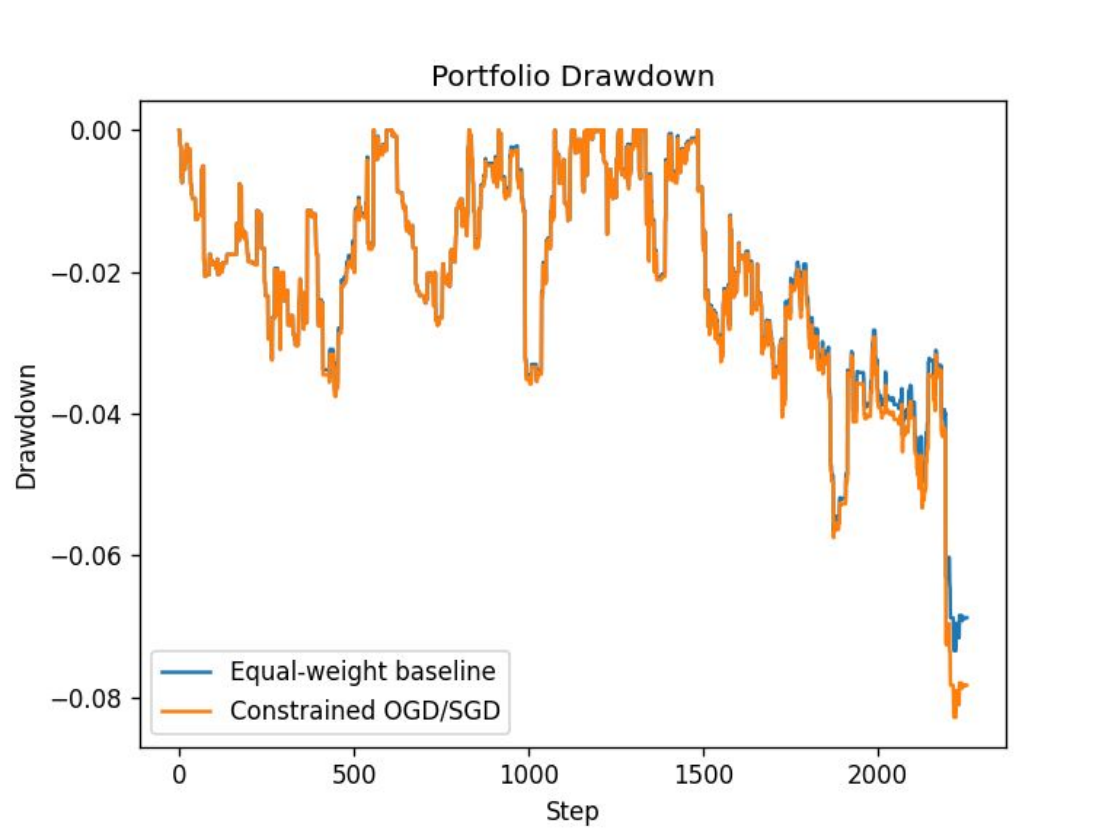
Equity Comparison

- The constrained strategy performs almost identically to the equal-weight baseline, with no clear outperformance.
- Adding domain and weight constraints preserves stability but does not meaningfully improve returns.



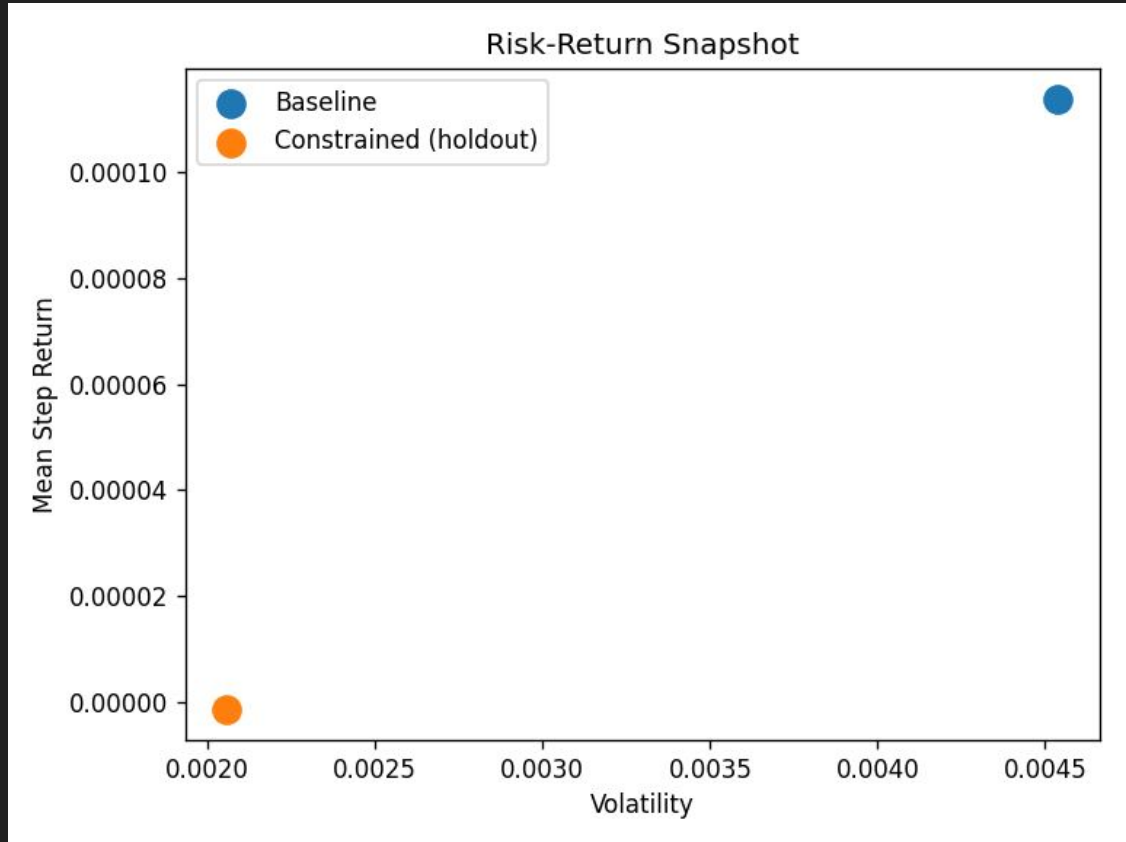
Drawdown Comparison

- Both strategies experience very similar drawdown patterns throughout the sample.
- The constrained portfolio does not meaningfully reduce max drawdown relative to the baseline.



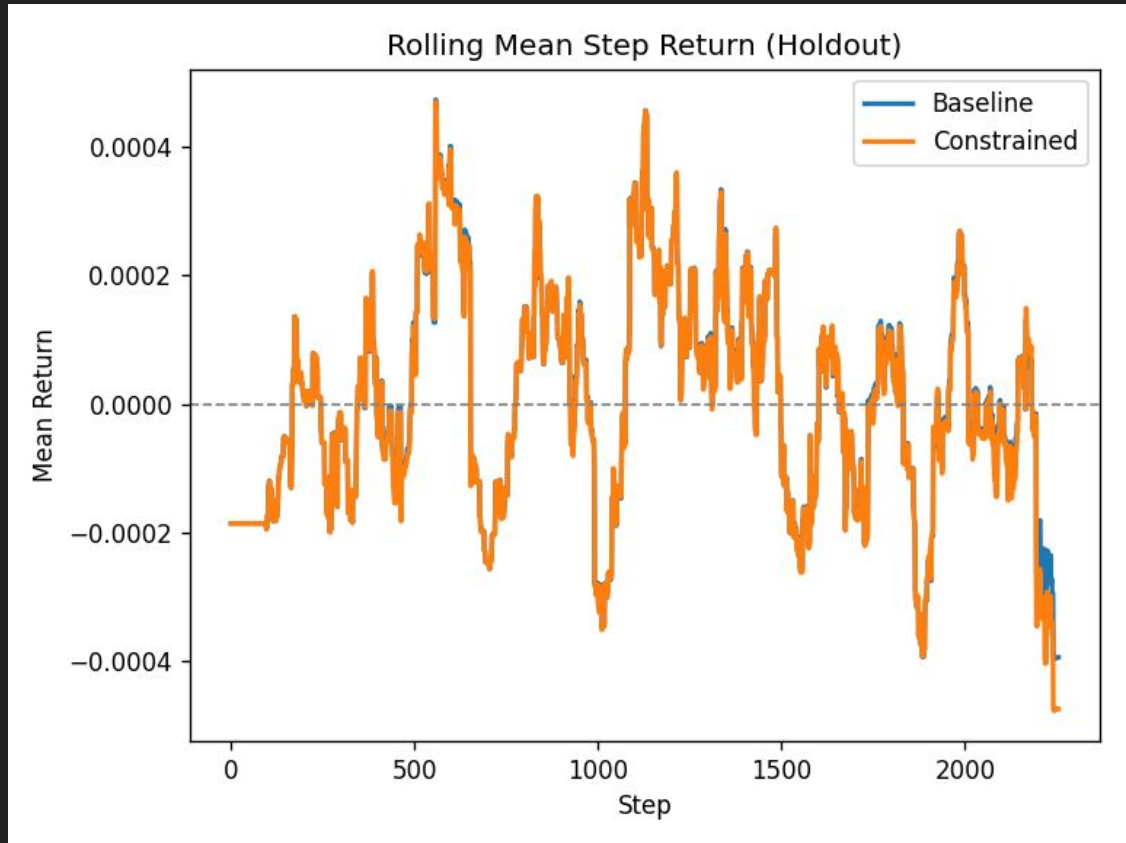
Risk-Return

- The constrained portfolio has lower volatility but also near-zero mean return in the holdout.
- The baseline delivers higher return with higher risk, resulting in a better risk-return tradeoff here.



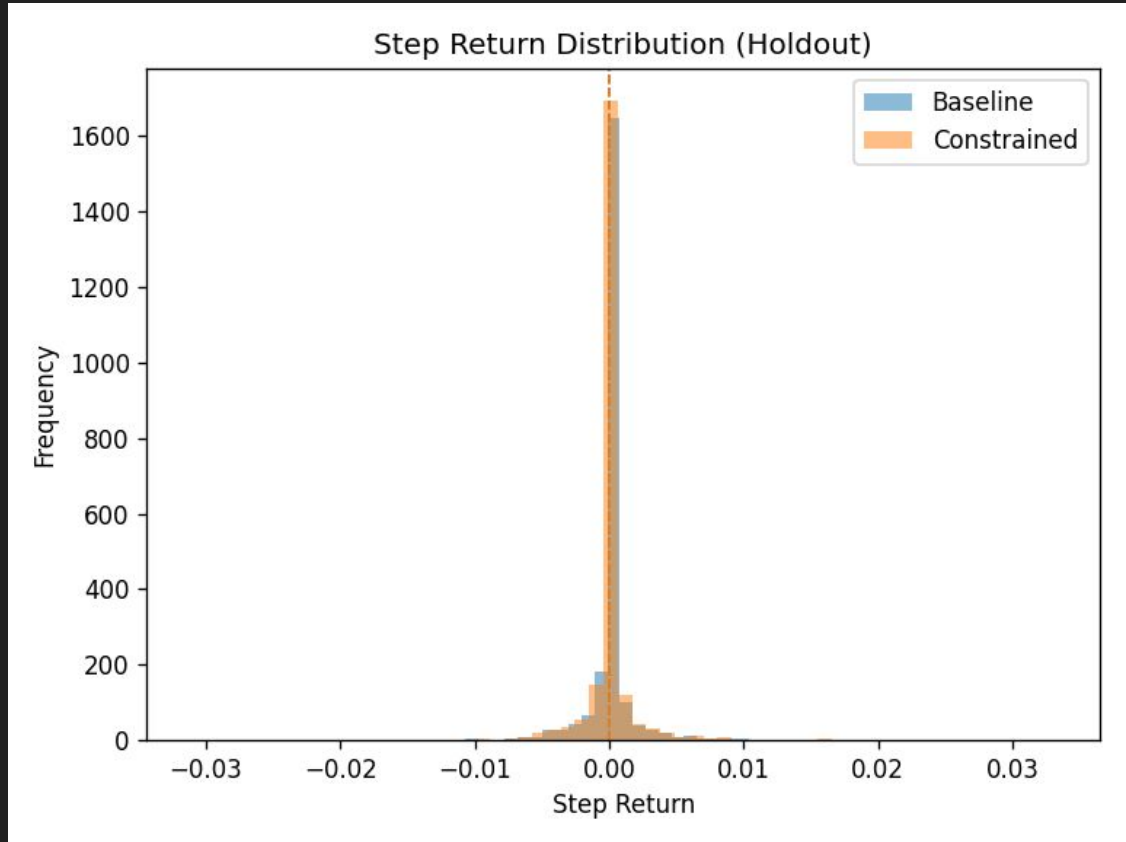
Rolling Mean Return Comparison

- Both strategies show highly unstable rolling mean returns that frequently cross zero.
- The constrained portfolio does not produce a more persistent positive return pattern in holdout.

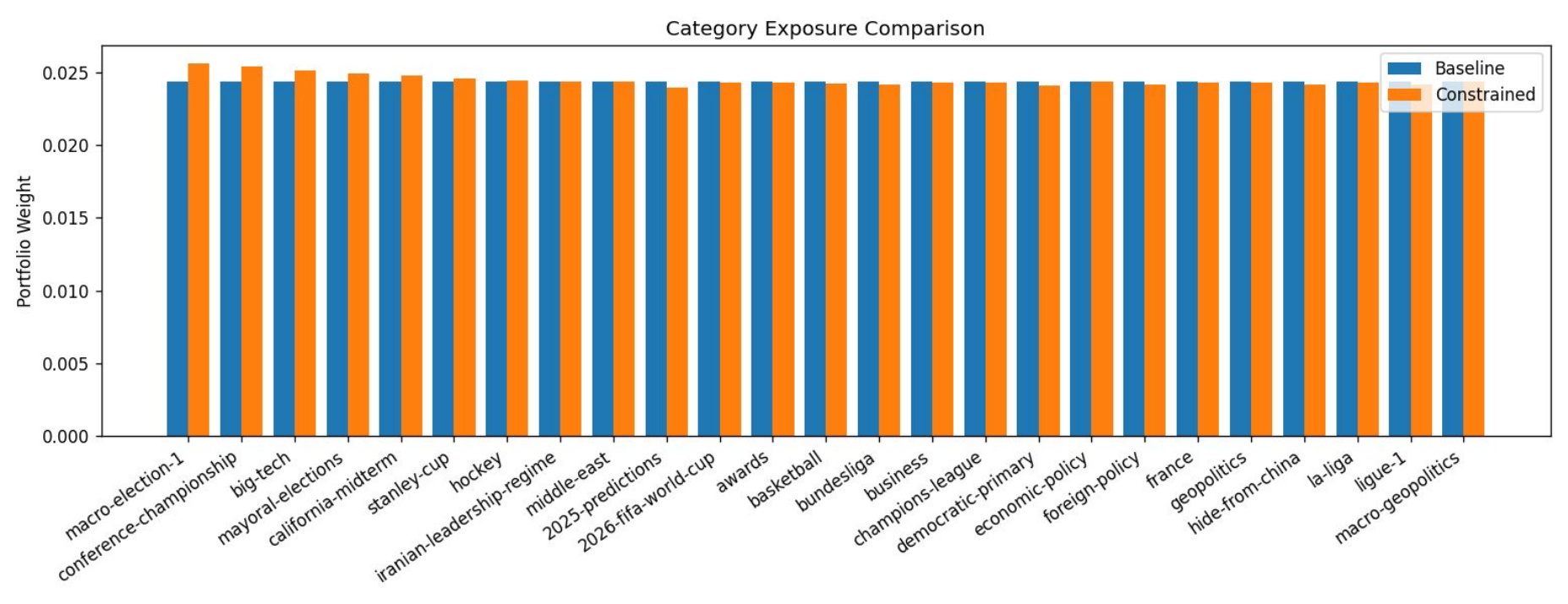


Return Distribution Comparison

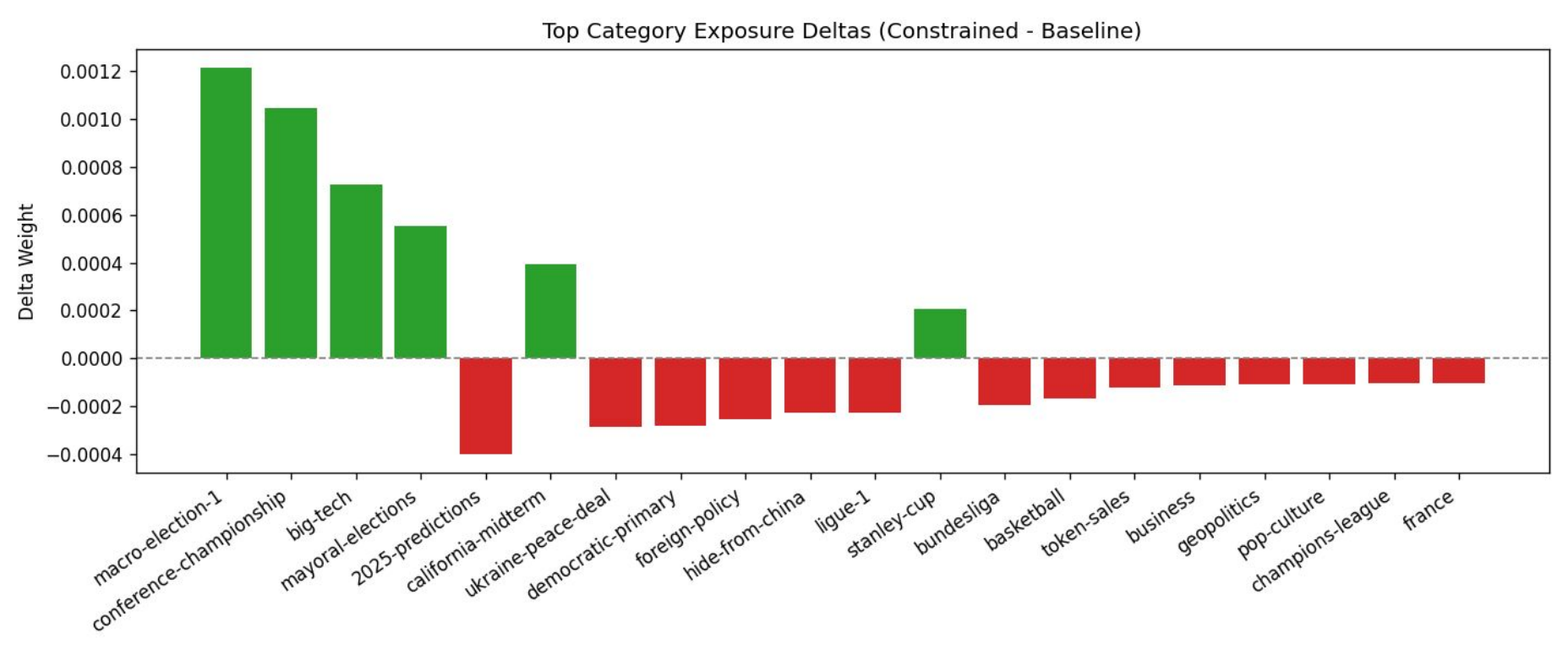
- Both strategies have highly concentrated return distributions centered near zero.
- The constrained portfolio slightly tightens dispersion but does not materially shift the mean return.



Exposure Results (1/2)



Exposure Results (2/2)



Current Progress

- Markets retained: 41
- Category count: 41
- Current constrained setup: `uniform_mix=0.88,max_weight=0.05,domain_limit=0.12`
- Tuning (walk-forward) constrained Sortino: 0.0517
- Holdout constrained mean return: -1.38e-06 (near flat)
- Holdout constrained max drawdown: -8.29%
- Baseline Sortino: 0.0466
- Baseline max drawdown: -7.84%